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National Institute of Business Management
Kandy

Higher National Diploma in Software Engineering

HNDSE 25.1F

AgroSmart: -

An IoT-Based Smart Agriculture Monitoring System
with
Multi-Sensor Control

Internet of Things – Proposal

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Date of Submission – 11/01/2026

Declaration

We, the undersigned, hereby declare that this proposal report for AgroSmart: IoT-Based Smart Agriculture Monitoring System with Multi-Sensor Control has been prepared by us (**Group 01**) as part of a group coursework for the Internet of Things module at the National Institute of Business Management, Kandy.

We confirm that:

1. The work presented is our own and has not been submitted previously for any other academic award.
2. All sources of information, ideas, and data from other authors have been properly cited and referenced.
3. The project is carried out in accordance with the institution's academic integrity policies.

We take full responsibility for the content and authenticity of this report.

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Abstract

This project, **AgroSmart: IoT-Based Smart Agriculture Monitoring System with Multi-Sensor Control**, presents the development of an IoT-based agriculture monitoring system for real-time observation and automated irrigation control. The system collects environmental data such as temperature, humidity, soil moisture, and light intensity using sensors interfaced with an ESP32 microcontroller. The collected data is transmitted to a cloud-based platform using Firebase Realtime Database and is visualized through a web-based dashboard and a mobile application, enabling remote monitoring of agricultural conditions.

A smart irrigation mechanism is implemented using a relay-controlled water pump, which operates automatically based on soil moisture levels to optimize water usage. The system is designed for a specific geographical region, initially focusing on the Kandy area in Sri Lanka, and follows a modular architecture to allow future enhancements such as additional sensors and advanced data analytics. This project is carried out as a group coursework and demonstrates the practical application of IoT and cloud technologies in modern smart agriculture systems.

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Table of Contents

Declaration ii

Abstract iii

Acknowledgement iii

Executive Summary 1

Project Background 2

Project Aim and Objectives 3

Project Aim 3

Project Objectives 3

Primary Objectives 3

Secondary Objectives 4

Scope of the Project 5

Methodology 6

System Phases 6

1. Hardware Layer: 6

2. Data Collection: 7

4. Data Description: 8

5. Dashboard and Mobile Application: 8

6. Testing & Validation: 8

7. AI-Based Smart Precision System .... **Error! Bookmark not defined.**

Tools and Technologies 8

Project Deliverables 9

Timeline 10

References 11

Appendices 12

Executive Summary

AgroSmart is an IoT-based Smart Agriculture Monitoring System designed to support farmers by providing real-time monitoring and automated control of key environmental parameters in agricultural fields. The system monitors temperature, humidity, soil moisture, and light intensity using a network of sensors interfaced with an ESP32 microcontroller. The collected sensor data is transmitted to a cloud-based platform using Firebase Realtime Database and is visualized through a web-based dashboard and a mobile application, enabling remote access and real-time observation of field conditions.

In addition to monitoring, AgroSmart implements a smart irrigation system using a relay-controlled water pump that automatically operates based on soil moisture levels, helping optimize water use and reduce manual intervention. The system is developed for a specific geographical region, initially focusing on the Kandy area in Sri Lanka, and follows a modular architecture that allows future expansion with additional sensors, data analysis capabilities, and intelligent decision-support features.

The expected outcomes of the project include:

- Real-time environmental data collection and monitoring.
- Cloud-based data storage and visualization through web and mobile dashboards.
- Automated irrigation control based on soil moisture conditions.
- Comprehensive academic documentation, including reports and presentations, as part of a group coursework project.

Project Background

Agriculture in regions such as Kandy, Sri Lanka, faces several challenges, including inefficient water usage, changing environmental conditions, and limited access to real-time field data. Traditional farming practices largely depend on manual observation and experience, which makes it difficult to accurately monitor parameters such as soil moisture, temperature, humidity, and light intensity. As a result, irrigation is often inefficient, leading to water wastage and reduced crop productivity.

Recent advancements in IoT and cloud computing technologies have made it possible to continuously monitor environmental conditions in agricultural fields and access this information remotely. By deploying sensor-based monitoring systems, farmers can gain real-time visibility into field conditions and make more informed decisions regarding irrigation and resource management. Cloud platforms further enable secure data storage and real-time visualization through dashboards and mobile applications.

The motivation behind the AgroSmart project is to develop an academic-level IoT-based smart agriculture monitoring system that demonstrates the practical application of modern technologies in agriculture. The system provides a foundation for efficient environmental monitoring and automated irrigation while maintaining a modular design that supports future enhancements such as advanced data analysis and intelligent decision-support features.

Project Aim and Objectives

Project Aim

The AgroSmart project aims to develop an IoT-based smart agriculture monitoring system that collects real-time environmental data and supports automated irrigation to improve efficient resource usage.

Project Objectives

Primary Objectives

1. To design and implement an IoT-based agriculture monitoring system using multiple sensors, including:
    - Temperature & Humidity Sensor (DHT11).
    - Soil Moisture Sensors (Capacitive & Resistive).
    - Light Intensity Sensor (LDR/MH-LDR).
    - Ultrasonic Sensor for water level detection.
    - Relay-controlled pump for smart irrigation.
2. To continuously collect real-time environmental data from the deployed sensors and securely store the data on a cloud-based platform (Firebase Real-time Database)**.**
3. To develop a web-based dashboard and a mobile application for real-time visualization and monitoring of environmental sensor data.
4. To implement an automated irrigation control mechanism based on soil moisture levels using predefined decision logic to optimize water usage.
5. To design the system for a specific geographical region, initially focusing on the Kandy area in Sri Lanka, while maintaining adaptability to other regions.
6. Integrate **AI/ML algorithms** to provide:
    - Irrigation recommendations based on humidity and soil moisture.
    - Crop-specific growth suitability analysis,
    - Seasonal climate trends and predictive insights.
7. To support automated irrigation control based on analyzed sensor data and predefined decision logic
8. To ensure a modular and scalable system architecture that allows future integration of additional sensors, analytical features, and system enhancements.

Secondary Objectives

9. To collect and analyze historical environmental and crop-related datasets from external sources to understand climate patterns and agricultural conditions.
10. To extend the system with data analysis and intelligent decision-support features based on collected sensor data and historical datasets.
11. To support crop-focused analysis for a selected set of crops by evaluating environmental conditions and seasonal trends.
12. To enable future expansion of the system through the integration of additional sensors, crops, dashboards, or advanced analytical modules while maintaining system reliability and scalability.

Scope of the Project

1. Target a specific area: **Kandy, Sri Lanka**, for initial implementation.
2. Monitor environmental parameters: **temperature, humidity, soil moisture, light intensity,** and **water levels.**
3. Provide actionable insights for farmers, including:
    - Optimal irrigation schedule.
    - Crop selection recommendations.
    - Predictions of climate and soil trends.
4. Modular and expandable system for additional sensors or AI modules.
5. Academic scope: real-time IoT system and AI-based recommendation system.
6. Focused on **proof-of-concept**, not large-scale industrial deployment.

Methodology

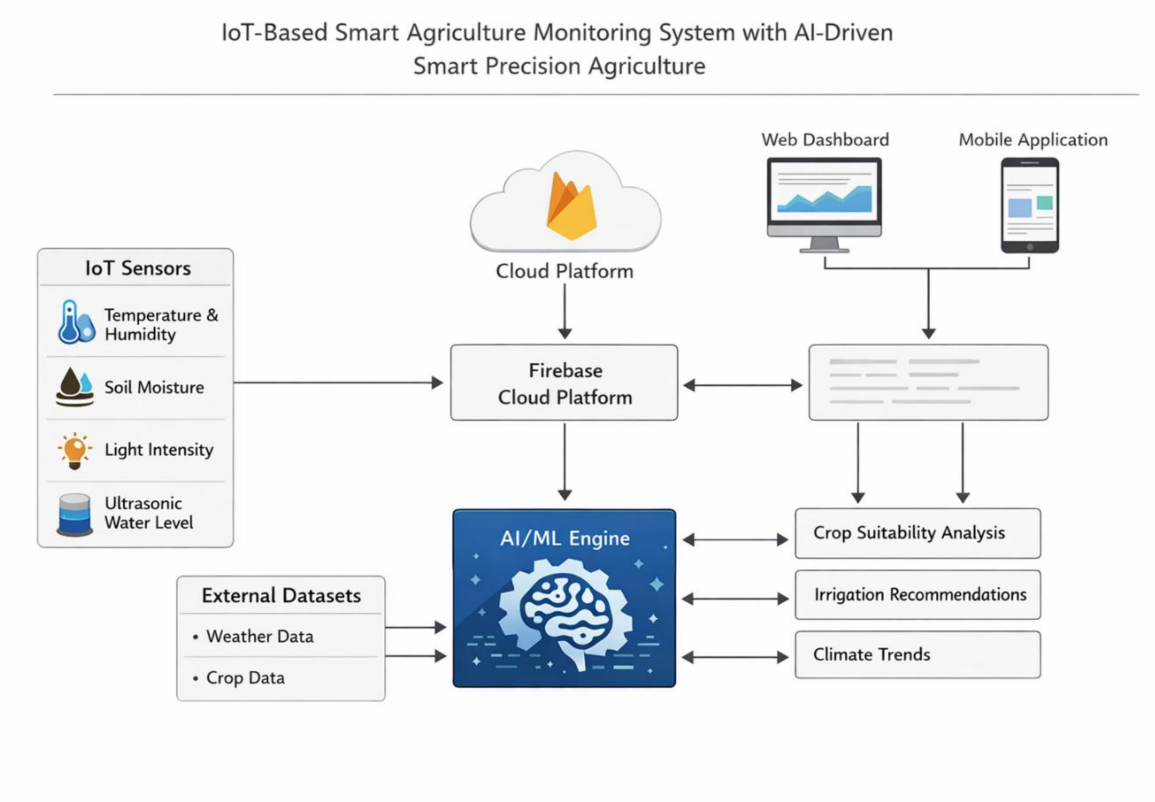


Figure 1: Overall Architecture of AgroSmart System

System Phases

1. Hardware Layer:

- ESP32 microcontroller as the main controller
- Sensors: DHT11, soil moisture, LDR, ultrasonic
- Relay module and water pump for automated irrigation
- Proper power management: 3.3V for low-voltage sensors, 5V external supply for pumps, and high-voltage sensors

2. Data Collection:

- Sensors collect real-time environmental data
- ESP32 uploads data to **Firebase Realtime Database**
- A dataset for the historical data

Sensor Name	Parameter Measured	Unit	Location	Purpose
DHT11	Temperature, Humidity	°C, %	Field environment	To monitor ambient environmental conditions affecting crop growth
Soil Moisture Sensor	Soil Moisture Level	Percentage (%)	Soil near plant root zone	To determine soil water content and enable automated irrigation decisions
Ultrasonic Water Level Sensor	Water Level	Centimeters (cm)	Water tank / reservoir	To monitor available water level for irrigation management
LDR Light Sensor	Light Intensity	Lux	Field / crop canopy area	To assess sunlight availability for photosynthesis analysis

Figure 2: Sensor Data Collected by AgroSmart

3. Data Analysis & AI/ML Module:

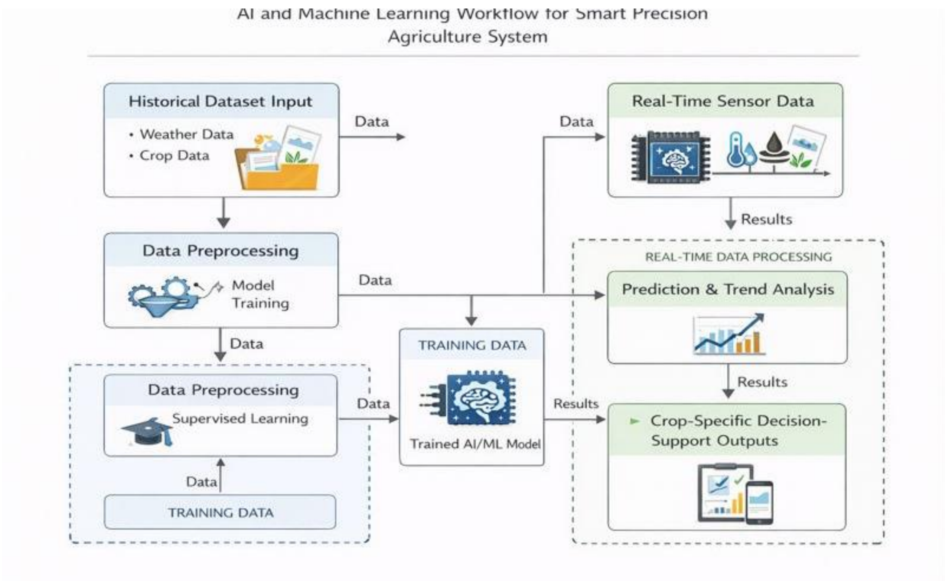


Figure 3: Data Flow from Sensors to Cloud and AI Engine

- Use of external datasets (Kaggle, public sources)
- Learning environmental patterns
- Crop-specific analysis
- Decision-support outputs

- AI/ML approach:
    - Supervised learning for historically labeled data is available
    - Unsupervised learning for pattern detection in unlabeled data

4. **Data Description:**
    - **Dataset data** (historical weather, crop data)
    - **Real-time sensor data**
    - Data types:
        - Numerical
        - Time-series
        - Labeled

5. **Dashboard and Mobile Application:**
    - Visualize sensor readings in real-time
    - Display recommendations and alerts
    - Allow remote monitoring of irrigation status

6. **Testing & Validation:**
    - Test sensors individually and in combination
    - Calibrate soil moisture thresholds and water tank levels
    - Validate AI predictions using sample datasets

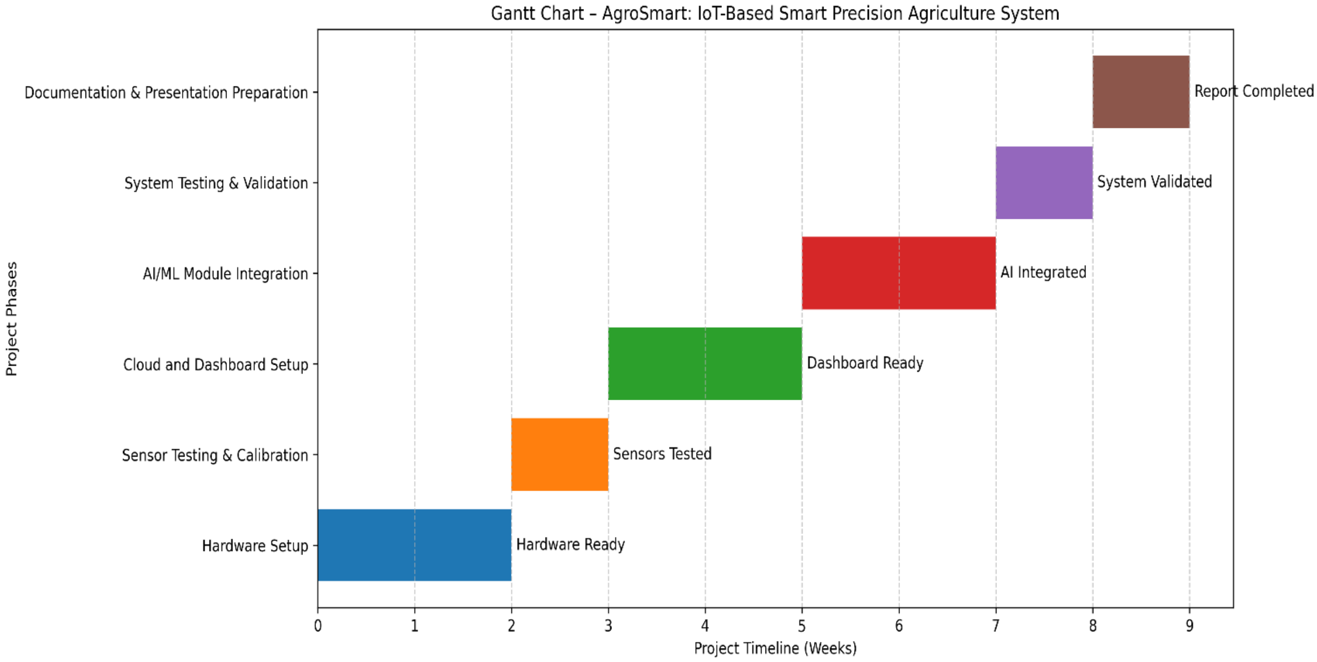
Tools and Technologies

- Microcontroller: ESP32
- Sensors: DHT11, Capacitive/Normal Soil Moisture, LDR, Ultrasonic
- Cloud Platform: Firebase Realtime Database
- AI/ML: Python, TensorFlow / scikit-learn
- Dashboard / App: Web technologies (HTML/CSS/JS) or Flutter for mobile
- Version Control & collaboration: GitHub - Cloud-based Version Control System

Project Deliverables

1. **Hardware Prototype:** ESP32 + sensor network + relay/pump system
2. **Software Module:** Cloud-based data storage, web dashboard, mobile app
3. **AI/ML Module:**
    - Crop suitability analysis
    - Trend prediction outputs
    - AI-generated insights visualized on the dashboard
4. **Reports & Documentation:** Proposal report, final report
5. **Presentations:** Proposal, progress updates, final presentation

Timeline



Phase	Duration
Hardware setup	2 weeks
Sensor testing & calibration	1 week
Cloud and dashboard setup	2 weeks
AI/ML module integration	2 weeks
System testing & validation	1 week
Documentation & presentation preparation	1 week

References

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- Zhang, L., & Kumar, A. (2020). *AI-driven smart irrigation systems: A review*. Agricultural Engineering Journal, 15(2), 110–125.

Appendices

- **Hardware Block Diagram:** ESP32 connected to sensors, relay, and pump
- **System Architecture Diagram:** Data flow from sensors → ESP32 → Firebase → AI module → Dashboard
- **Sample Sensor Data Table:** Example readings for temperature, humidity, soil moisture, and light intensity
- **Additional Figures/Charts:** Sensor calibration charts, optional images of sensors